Housing Discrimination and Algorithmic Bias: A Data Ethics Analysis of Hennepin and Ramsey County, Minnesota | DATA 6550

Algorithmic Bias

Group 3 - Data 6550 - Spring 2026
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Note: AI agents such as ChatGPT were utilized in the course of this project for coding help and some suggestions on where to focus analysis.

1. Introduction and Historical Context:

The economic and demographic aspects evident in many American towns have been primarily ascribed to housing discrimination. Racial covenants were one of the most prominent ways that housing discrimination was enforced. Legal stipulations in deeds known as "racial covenants" ban members of a given racial group from selling a property. This sort of discrimination, which banned non-white individuals from owning property in specific communities, was popular in the early and mid-20th centuries. The Fair Housing Act of 1968 rendered racial covenants illegal, however their influence may still be seen in modern neighborhoods. Restricted communities have traditionally experienced more investments, superior infrastructure, and longer-term gains in property prices. However, there might have been fewer investment chances in the form of homeownership for populations that were restricted from acquiring and owning property in certain places. These days, computer algorithms that analyze risks using financial and geographical data to determine whether to accept or deny a loan are the key factor determining mortgage lending selections. Historical inequities may show as biases in data or algorithmic prejudice in assessing loan applications, even when race is not a part in those selections. The goal of this study is to evaluate the effect of previous housing discrimination and lending practices in two Minnesota counties, including algorithmic and data bias. The historical racial covenants in the two Minnesota counties under investigation, Hennepin and Ramsey, are depicted in red on the maps below. Appendix B offers maps that demonstrate the historical covenants.

Dataset	**Records**	**Coverage**
mortgage_covenant_data.csv	100,931 applications	Minnesota statewide HMDA + Hennepin covenant density
covenants-mn-ramsey-county.csv	5,528 covenant records	Ramsey County, 1913–1961
county_27123.csv	14,482 applications	Ramsey County HMDA 2023 (full)

2. Methodology:

Two kinds of bias were investigated:

• Data bias, or historical prejudice engrained in modern elements like neighborhood demographics and property pricing

• Algorithmic bias: modern lending algorithms that employ proxy criteria to achieve racially disproportionate outcomes

2.1 Data Sources

Source	**File**	**n**	**Key Variables**
HMDA 2023 (Hennepin+MN)	mortgage_covenant_data.csv	100,931	Race, loan amount, approval, covenant density, interest rate, income
Mapping Prejudice (Ramsey)	covenants-mn-ramsey-county.csv	5,528	Deed year, city, zip, covenant language
HMDA 2023 (Ramsey)	county_27123.csv	14,482	Race, approval, DTI, LTV, interest rate, tract demographics

2.2 Variable Definitions

was_approved: This is true if action_taken = 2 (approved, not accepted) or 1 (loan began). Every other HMDA code is false.

Covenant_density: Low (1–50), Medium (51–100), High (101–200), and Very High (>200 covenants) are categories that were pre-joined to HMDA at the census-tract level. Covenant-free tracts constitute the reference group without a covenant.

tract_minority_population_percent: The percentage of non-white residents in each census tract as assessed by the ACS. In Ramsey County, when tract-level covenant links were unavailable, it operated as a stand-in for covenant effect.

2.3 Analytical Approach

Four steps were involved in the analysis: (1) a descriptive comparison of the characteristics of covenanted and non-covenanted tracts; (2) Welch two-sample t-tests for statistical significance; (3) logistic regression to find proxy variables influencing approval decisions; and (4) tract-level stratification in Ramsey County using minority population quartiles as a covenant proxy. Python 3.11 was used for all analysis, along with pandas, scipy, scikit-learn, matplotlib, and seaborn.

3. Findings

3.1 Data Bias -Property Value Disparities

The most surprise outcome is that historically covenanted plots had larger property values. In covenanted tracts, the typical property value is $355,000, but in non-covenanted plots, it is $315,000. This difference of $40,000 is statistically significant ($t = 7.61$, $p < 0.0001$). Once the process is established, this is not counterintuitive: developers purposely applied racial covenants to well-funded, attractive suburbs in order to insulate them from integration. well well over 70 years, such regions experienced sustained investment and appreciation.

Table 1. Median property values rise monotonically with covenant density. t = 7.61, p < 0.0001.

Covenant Density	Median Property Value	vs. Non-Covenanted
Low (1–50)	$325,000	+$10,000
Medium (51–100)	$365,000	+$50,000
High (101–200)	$375,000	+$60,000
Very High (>200)	$385,000	+$70,000
Non-Covenanted	$315,000	Reference

3.2 Data Bias - Neighborhood Demographics

In 2023, covenant geography continues to impact who lives where. In covenanted tracts, the average minority population percentage is 30.4%, compared to 21.9% in non-covenanted tracts. The border zones where covenant enforcement was contested or inadequate areas that typically underwent racial change after 1968 are illustrated by low-density covenanted tracts (37.5% minority). Due to the strongest implementation of exclusion, particularly high density tracts (25.4%) remained more totally white.

3.3 Data Bias - Economic Indicators

Covenant density also boosts the median applicant income: Low = $100,000, Medium = $112,000, High = $117,000, Very High = $128,000—all of which are higher than the $94,000 median for non-covenanted tracts. Decades of wealth concentration are evident in the $17,000 income premium in covenanted zones. An algorithm is partly employing a variable produced by racially discriminatory investing patterns when it utilizes income as an approval factor.

3.4 Algorithmic Bias - Racial Approval Rate Patterns

The Hennepin dataset's overall mortgage approval rate is 66.47%. When broken down by race, a clear hierarchy with a Black-White discrepancy of 15.2 percentage points is revealed:

Race	Approval Rate	Count
White	75.6%	63,627

Asian	70.8%	5,905
American Indian or Alaska Native	62.2%	709
Black or African American	60.4%	5,199
Native Hawaiian / Pacific Isl.	56.6%	113

Table 2. Mortgage approval rates by race, Hennepin County HMDA 2023.

Crucially, historically covenanted neighborhoods show a bigger Black-White difference (18.5 percentage points) than those outside (14.7 percentage points). Historical geography exacerbates racial imbalances even within the same algorithm, making candidates from covenanted neighborhoods even more disadvantaged.

Tract Type	White Approval	Black Approval	Gap
Covenanted tracts	76.7%	58.2%	18.5 pp
Non-covenanted tracts	75.5%	60.7%	14.7 pp

Table 3. Black-White approval gap is 3.8pp larger in historically covenanted areas.

3.5 Algorithmic Bias - Loan Term Differences

Interest rates vary even among borrowers who have been approved. Covenanted tracts have a mean interest rate of 6.610%, but non-covenanted tracts have a mean interest rate of 6.571%. This difference of 0.040 percentage points is statistically significant ($t = 2.79$, $p = 0.005$). This amounts to an additional $4,183 in interest paid over the duration of a $350,000 30-year mortgage.

3.6 Algorithmic Bias - Risk Assessment Criteria and Proxy Variables

The criteria that most effect lending choices are indicated by a logistic regression model predicting mortgage approval ($n = 67,070$, accuracy = 90.3%):

Variable	Coefficient	Direction	Bias Implication
property_value	+1.54	Positive	Proxy for covenant history -strongest driver
loan_amount	-1.27	Negative	Financial risk factor
income	+0.53	Positive	Financial factor; covenanted areas earn more
tract_to_msa_income_pct	+0.29	Positive	Neighborhood wealth - correlated with covenants
interest_rate	-0.12	Negative	Financial risk factor
tract_minority_pop_pct	-0.077	Negative	Direct algorithmic bias - race proxy
covenant_count	-0.021	Negative	Historical bias signal -still predicts denial

Table 4. Logistic regression coefficients. Negative minority_pop_pct and covenant_count confirm algorithmic bias.

At +1.54, which is nearly twice as high as the next variable, **property_value** is the single strongest predictor of acceptance. The algorithm inherits racial history via a variable that looks to be financially neutral when property_value is utilized as the principal underwriting input as property prices in covenanted zones are continuously higher due to more than 70 years of racially limited investment. The dataset's most evident proxy variable is this one.

Even after correcting for income, loan amount, property value, and interest rate, **tract_minority_population_percent** had a coefficient of −0.077. Living in a census tract with a bigger minority population diminishes the chance of acceptance regardless of socioeconomic circumstances. This is a clear, tangible indicator that the algorithm considers the racial mix of the neighborhood as a risk factor.

3.7 County Comparison - Ramsey County Extension

We broaden the investigation and confirm the same structural tendencies using two particular Ramsey County datasets.

Covenant History (covenants-mn-ramsey-county.csv): From 1913 to 1961, there were 5,528 documented racial covenants in Ramsey County, predominantly in Saint Paul (2,566) and the inner suburbs of Roseville (668), Falcon Heights (605), and Mounds View (515). With 2,304 recordings, the 1940s were the leading decade. This happened during the post-World War II suburban housing boom, when returning soldiers purchased properties in freshly developed, covenant-protected suburbs.

Decade	Covenants Recorded	Context
1910s	349	Early suburban development
1920s	1,422	Rapid suburban expansion
1930s	965	Depression-era, continued enforcement
1940s	2,304	Peak post-WWII suburban boom
1950s	480	Declining as FHA began pushing back
1960s	8	Near end Fair Housing Act 1968

Table 5. Ramsey County racial covenants by decade. Source: covenants-mn-ramsey-county.csv.

Modern Lending (county_27123.csv): 14,482 mortgage applications from 142 census tracts make up the complete 2023 Ramsey County HMDA dataset. Ramsey's bigger minority population (40.2% vs. 22.9% statewide) is reflected in the overall approval rate of 60.8%, which is lower than Hennepin's 66.47%. There is a 16.9 percentage point disparity between Black and White approval.

Metric	Ramsey County	Hennepin (Covenanted)
Total Applications	14,482	11,169
Overall Approval Rate	60.8%	67.1%
White Approval Rate	67.5%	76.7%
Black Approval Rate	50.6%	58.2%
Black-White Gap	16.9 pp	18.5 pp
Median Property Value	$315,000	$355,000
Mean Interest Rate	6.879%	6.610%
Mean Minority Pop %	40.2%	30.4%
Mean DTI Ratio	43.0%	Not available
Documented Covenants	5,528	30,000+
Covenant Year Range	1913-1961	1910-1955

Table 6. County comparison - Ramsey vs Hennepin. Sources: county_27123.csv and mortgage_covenant_data.csv.

Tract-Level Proxy Analysis (Ramsey): We employ tract_minority_population_percent as a covenant proxy—the same approach proven in Hennepin—because the Ramsey covenant CSV cannot be directly connected to HMDA at the tract level without a GIS spatial join (see Section 5). Historically covenanted suburban areas are represented by low-minority tracts (bottom quartile, ≤21.3%), while traditionally excluded urban neighborhoods are represented by high-minority tracts (top quartile, ≥59.3%).

Tract Type (Ramsey)	Approval Rate	White	Black	Gap	Median Prop. Value
Low Minority (≤21.3% — likely covenanted)	64.5%	69.8%	50.0%	19.8 pp	$385,000
High Minority (≥59.3% — historically excluded)	55.1%	60.9%	48.1%	12.8 pp	$255,000

Table 7. Within-Ramsey tract analysis mirrors Hennepin: larger gap in likely-covenanted areas.

The Hennepin finding is identically paralleled by the 19.8pp Black-White split in low-minority Ramsey tracts (presumably covenanted) versus 12.8pp in high-minority tracts (historically excluded), suggesting that the structural trend is constant across both counties regardless of the dataset employed.

4. Discussion

4.1 Consequences for Contemporary Lending

This study indicates that algorithmic mortgage underwriting systems are not neutral instruments; via ostensibly objective financial parameters, they inherit the racial geography generated by historical discrimination. The basic mechanism is simple: attractive areas were protected by racial covenants, which drew constant investment and boosted property prices. These property values were then employed as the major input into lending algorithms, which favor applications from traditionally white areas and penalize those from historically excluded communities without ever mentioning race. Because it continues after correcting for income, property value, loan amount, and interest rate, the negative coefficient on tract minority population % (−0.077) is particularly significant. This shows that the algorithm is intentionally penalizing applicants based on the racial mix of their location rather than passively responding to financial risk characteristics that just so happen to correlate with race. Regardless of aim, this is deemed disparate impact under the Fair Housing Act and the Equal Credit Opportunity Act.

4.2 Suggested Strategies for Mitigation

4.2.1 Handling Bias in Data:

Appraisal Reform: Apply upward adjustment factors for census tracts with established covenant history, or retrain computerized valuation models using only post-1968 data. The single greatest approval driver is property_value (+1.54); changing this input would have the most affect on fairness.

CRA Reinvestment Incentives: In order to offset the cumulative costs of disinvestment, the Community Reinvestment Act should give additional credit at affordable rates to lenders who are actively investing in previously excluded neighborhoods.

Audit Neighborhood Variables: There is a considerable association between racial composition and tract_to_msa_income_percentage (+0.29). Lenders want to study whether deleting or changing this element decreases racial inequities without substantially impacting model accuracy.

4.2.2 Handling Algorithmic Prejudice:

Counterfactual Fairness Testing: Set tract_minority_population_percent to zero and examine the model's findings for each application. The variable is serving as a race proxy and needs to be deleted or capped if the outcome changes.

Geographic Disparate effect Monitoring: Lenders are mandated by ECOA to keep a look out for disparate impact. The 3.8pp gap amplification inside covenanted areas would be evident and actionable if this monitoring explicitly tested by covenant location rather than merely applicant race.

comprehensive Ramsey Spatial Join: By employing a geopandas spatial join to match the Ramsey covenant data to census tracts, comprehensive tract-level covenant density analysis would be achievable, providing the same depth of monitoring in Hennepin and Ramsey.

5. Limitations and Future Work

5.1 Limitations

• **Missing Credit Score Information:** Individual credit scores, debt-to-income ratios, and down payment amounts are not included in the Hennepin HMDA dataset. DTI (mean = 43.0%, n = 5,265 of 14,482 entries) is contained in the incomplete Ramsey HMDA dataset (county_27123.csv). We cannot properly identify discrimination from unmeasured creditworthiness discrepancies without comprehensive credit data.

• **Ramsey Tract Join Not Completed:** There are no census tract IDs in covenants-mn-ramsey-county.csv, and there are no zip codes or parcel PINs in county_27123.csv. A Census-based geopandas spatial join
For a thorough tract-level covenant density study in Ramsey, TIGER/LINE shapefiles would be required. As a proxy, we chose tract_minority_population_percent, which is less exact but methodologically sound.
• **Race Self-Reporting:** Non-random missingness may bias estimates of racial disparities; 22,811 records (22.6%) in the Hennepin dataset and 3,112 records (21.5%) in the Ramsey dataset have "Race Not Available."
•**Single Year of Data:** The study is confined to 2023. We are unable to establish if the gaps are getting better or becoming worse over time.
• **Correlation, Not Causation:** We demonstrate that current loan results are associated with covenant history. Longitudinal data tying individual tracts to covenant records and loan histories spanning many decades would be essential to show causal causation.

5.2 Future Work

• To allow tract-level covenant density analysis that matches the Hennepin depth, finalize the geopandas spatial join for Ramsey County using Census TIGER/LINE shapefiles.
• Extend the study to the counties of Dakota, Washington, and Anoka, all of which had extensive racial covenant activity inside the metropolitan region of the Twin Cities.
• To establish if the Black-White approval disparity has been shrinking or rising over time, consider HMDA data from 2018 to 2022.
• Build a more complete logistic regression model that adjusts for creditworthiness using the full DTI and LTV data provided in county_27123.csv.
• Using geopandas and Folium, develop an interactive map that overlays covenant density with current approval rates to enable for visual examination of the regional trends observed in this investigation.

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